Some Notes:

i) Algebraical or trigonometrical transformations sometimes prove easier to perform a complicated differentiation.

Example 10.4:

a) Let $y = (cosecx)^{\cot x}$, then $logy = cotxlogcosecx$

Differentiating both sides with respect to x,

$$\frac{1}{y}\cdot\frac{dy}{dx} = -\text{cosec}^2 x \log \text{cosec } x + \frac{\cot x}{\text{cosec } x}(-\text{cos ec } x \cot x)$$

$$= -(\text{cosec}^2 x \log \text{cosec } x + \cot^2 x)$$

$$\therefore \frac{dy}{dx} = -(\text{cosec}x)^{\cot x}(\text{cosec}^2 x \log \text{cosec } x + \cot^2 x)$$

b) If $y = \tan^{-1}\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{\sqrt{1+\sin x}+\sqrt{1-\sin x}}$, then $y = \tan^{-1}\frac{1-\cos x}{\sin x}$

$$= \tan^{-1}\frac{2\sin^2\frac{1}{2}x}{2\sin\frac{1}{2}x\cos\frac{1}{2}x} = \tan^{-1}(\tan\frac{1}{2}x) = \frac{1}{2}x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2}$$

ii) Derivative of a determinant

Let

$$\Delta(x) = \begin{vmatrix} f_1(x) & \phi_1(x) & \psi_1(x) \\ f_2(x) & \phi_2(x) & \psi_2(x) \\ f_3(x) & \phi_3(x) & \psi_3(x) \end{vmatrix}$$

Then,

$$\Delta'(x) = \begin{vmatrix} f_1'(x) & \phi_1(x) & \psi_1(x) \\ f_2'(x) & \phi_2(x) & \psi_2(x) \\ f_3'(x) & \phi_3(x) & \psi_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & \phi_1'(x) & \psi_1(x) \\ f_2(x) & \phi_2'(x) & \psi_2(x) \\ f_3(x) & \phi_3'(x) & \psi_3(x) \end{vmatrix} +$$

$$\begin{vmatrix} f_1(x) & \phi_1(x) & \psi_1'(x) \\ f_2(x) & \phi_2(x) & \psi_2'(x) \\ f_3(x) & \phi_3(x) & \psi_3'(x) \end{vmatrix}$$

Example 10.5 :

$$\Delta(x) = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$$

$$\Delta'(x) = \begin{vmatrix} \cos x & \cos x & \sin x \\ -\sin x & -\sin x & \cos x \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} \sin x & -\sin x & \sin x \\ \cos x & -\cos x & \cos x \\ x & 0 & 1 \end{vmatrix} +$$

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & -\sin x & -\sin x \\ x & 1 & 0 \end{vmatrix}$$

$= \quad cosx\,(-sinx - cosx) - cosx\,(-sinx - cosx) + sinx\,(-sinx + sinx) + sinx(-cosx) + sinx(cosx - xcosx) + xsinxcosx + sin^2x - xsinxcosx + cosx\,(cosx + xsinx) = sin^2x + cos^2x = 1$

10.3 GEOMETRICAL INTERPRETATION OF DERIVATIVES

Geometrically, the derivative of a function is the slope of the concerned curve.

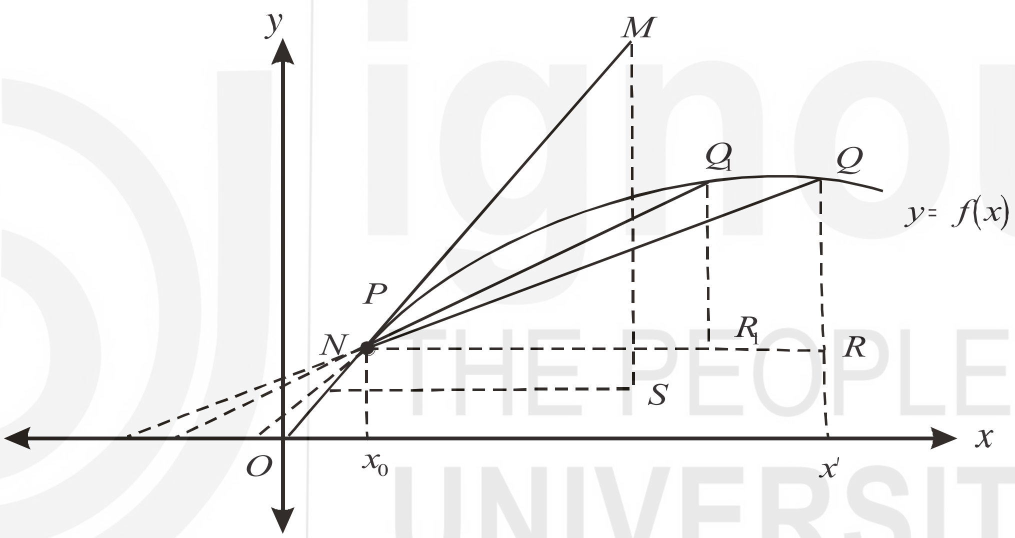

Fig.10.1: Graphical Representation of how ratio [f(x + h) – f(x)/h], represented by chords approaches the tangent, representing derivative f'(x).

The function drawn here is $y = f(x)$, let P and Q be the two points on this function. We start from the point P, where $x = x_0$ Suppose x has risen from x_0 to x', i.e., $\Delta x = x_0x'$. As a result, y has changed from P to Q on the curve and we get $\dfrac{\Delta y}{\Delta x} = \dfrac{QR}{PR} = \tan\phi$

This shows as Q approaches P, Δx increases and we get steeper and steeper lines, when $\Delta x \to 0$ the point Q almost coincides with P and the chord PQ ultimately becomes tangent at point P. Let MN be the tangent to $f(x)$ at point P. Thus, the derivative of a function at a point is actually the value of the slope of the function at that point. Thus, the derivative for a linear function is constant, but for a non-linear function the derivatives are different for different values of x.

10.4 DIFFERENTIALS

If $f'(x)$ is the derivative of $f(x)$, and Δx is an increment of x, then the differential of $f(x)$, denoted by the symbol $df(x)$, is defined by the relation $df(x) = f'(x)\Delta x$ … (10.1)

If $f(x) = x$, then $f'(x) = 1$ and (i) reduces to $dx = \Delta x$

Thus, when x is the independent variable, the differential of x $(= dx)$ is identical with Δx. Hence, if $y - f(x)$, then the relation (10.1) becomes $dy = f'(x)dx$ … (10.2)

i.e., the differential of a function is equal to its derivative multiplied by the differential of the independent variable.

Thus, if $y = tanx$, then $dy = sec^2 x dx$.

From the definition of the differential of a function, the following formulae are obvious –

$d(c) = 0$ where c is constant

$d(u + v - w) = du + dv - dw$

$d(uv) = udv + vdu$

$$\mathrm{d}\left(\frac{u}{v}\right) = \frac{v\mathrm{d}u - v\mathrm{d}u}{v^2}$$

Differentials are especially useful in applications of integral calculus.

Notes:

i) For the independent variable x, increment Δx and differential dx are equal, but same is not true with the dependent variable y, i.e., $\Delta y \neq dy$ generally.

ii) The relation (ii) can be written as $dy/dx = f'(x)$, i.e., the quotient of the differentials of y and x is equal to the derivative of y with respect to x.

10.5 HIGHER ORDER DERIVATIVES

The derivative of a function of x, $f(x)$, is in general a function of x. This new function may have a derivative, which is called the second derivative (or second differential coefficient) or, $f(x)$. The original derivative is called the first derivative (or, first differential coefficient). Similarly, the derivative of the second derivative is called the third derivative and so on for the n^{th} derivative.

Example 10.6: Let $y = 4x^5 + 7x^4 + 3x + 9$, then $\frac{dy}{dx} = 20x^4 + 28x^3 + 3$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = 80x^3 + 84x^2; \frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = 240x^2 + 168x$$

Check Your Progress 1

1) Find $\frac{dy}{dx}$ of the following:

i) $\mathrm{y} = \sin^2\mathrm{x} - \mathrm{e}^{\mathrm{x}} + 4^{\mathrm{x}}$

ii) $\mathrm{y} = 3\mathrm{x}^2\mathrm{cosecx}$

iii)$\mathrm{y} = \frac{\log \mathrm{x} - \sin \mathrm{x}}{2\mathrm{x}^2 - 1}$

iv) $\mathrm{y} = 2\cos \mathrm{v} + \tan^2\mathrm{v}, \ \mathrm{v} = \mathrm{e}^{3\mathrm{x}},$

v) $\mathrm{y} = \mathrm{a}\cos \Theta, \mathrm{x}\,\mathrm{x} = \mathrm{b}\sin \Theta$

....

....

....

2) Find $\frac{d^2y}{\mathrm{dx}^2}$ and $\frac{d^3y}{\mathrm{dx}^3}$ for the following functions:

$i) y = -e^x + x^4$

$ii) y = 3x^2 \log(2x)$

....

....

....

3) For natural number n, find n[th] derivative of each of the following functions

(i) y = e^x (ii) y = sin x (iii) y= cos x (iv)y =1/(1+x) (v) y= log (1+x)

....

....

....

10.6 APPLICATION OF SIMPLE DERIVATIVES IN ECONOMICS

The derivatives have wide application in economics. We always deal with functional relations while studying economic problems. While dealing with these functional relations, sometimes we get prompted to find out the rate of change of one (dependent) variable owing to the change of the other (independent) variable.

10.6.1 Average and Marginal Revenue

The price (p) set by the sellers in an imperfect market depends on the quantity of output (q) the sellers expect to sell, i.e., $p = f(q)$. This is the inverse demand function. The total revenue (R) obtained by the seller is $R = p.q = f(q).q$. Thus, R is a function of q only, i.e., $R = \phi(q)$. This implies R varies in response to a variation in q. We get two concepts – average revenue and marginal revenue. The average value of R, the average (AR) measures the average variation of R over a range of values of q extending from zero upto a certain specified value. The specified value of q is the one at which we seek the average value of R.

Thus, $\mathrm{AR} = \frac{\mathrm{R}}{\mathrm{q}} = \frac{\mathrm{pq}}{\mathrm{q}} = \mathrm{p} = f(\mathrm{q})$

Hence, the function $p = f\ (q)$ gives us the AR curve when plotted graphically. The marginal value of R or, the marginal revenue (MR) measures the variation of R at the margin.

Thus, $\mathrm{MR} = \frac{\mathrm{dR}}{\mathrm{dq}} = f(\mathrm{q}) + \mathrm{q}.f'(\mathrm{q})$

We can establish a relation between MR and AR: $MR = AR + q.f'(q)$

The relation describes that the difference between MR and AR will always be equal to $qf'(q)$. Since q is always non-negative and since $f'(q) < 0$, we have $MR - AR < 0$ i.e., $MR < AR$ for all q.

Thus, when the AR curve is downwards sloping, the MR curve always lies below the AR curve. However, if AR curve be upward rising $MR > AR$ and the MR curve will lie above the AR curve. When $f'(q) = 0$, the MR curve and the AR curves coincide. This happens when p is a given quantity and fixed for all q. The AR curve will then be a parallel line to the horizontal axis. When $q = 0$, $AR = f(0)$ and $MR = f(0)$. Thus, at $q = 0$ the two curves intersect each other.

10.6.2 Average and Marginal Cost

The total cost (c) of a firm depends on the total output (q) produced,

i.e., $c = c(q)$.

Proceeding by the same argument in the last section, average cost $(AC) = \frac{c(q)}{q}$ and marginal cost$(MC) = \frac{dc(q)}{dq}$.

Again, we can establish a relation between AC and MC. Let us differentiate AC function with respect to q.

$$\frac{\mathrm{d}}{\mathrm{dq}}\left(\frac{\mathrm{c}}{\mathrm{q}}\right) = \frac{\frac{\mathrm{dc}}{\mathrm{dq}}.\mathrm{q} - \mathrm{c}.1}{\mathrm{q}^2} = \frac{1}{\mathrm{q}}\left(\frac{\mathrm{dc}}{\mathrm{dq}} - \frac{\mathrm{c}}{\mathrm{q}}\right) = \frac{1}{\mathrm{q}}(\mathrm{MC} - \mathrm{AC})$$

If AC curve slopes downwards, then $\frac{\mathrm{d}}{\mathrm{dq}}\left(\frac{\mathrm{c}}{\mathrm{q}}\right) < 0$ or, $\frac{1}{\mathrm{q}}(\mathrm{MC} - \mathrm{AC}) < 0$ or, $MC < AC$.

i.e., MC curve lies below the AC curve when AC is falling. When AC slopes upwards, we can conclude MC curve lies above the AC curve. The AC curve intersects the MC curve if the slope of the AC curve is zero.

10.6.3 Elasticity of Demand

Demand is a multivariate function. We simplify our analysis by assuming demand to depend mainly on two factors – price and income. If one or both of these factors change, demand changes accordingly. The elasticity of demand measures the percentage change in quantity demanded due to one percent change in one of these two variables.

Suppose, the demand function is given by $q = f(p)$ where p denotes price and q quantity demanded. Then by definition, price elasticity of demand (e_d) is given by

$e_d = \frac{\frac{\Delta q}{q}\times 100}{\frac{\Delta p}{p}\times 100} = \frac{\Delta q}{\Delta p}.\frac{p}{q}$, where $\Delta p$and $\Delta q$are the changes in quantity demanded and price respectively. The price elasticity of demand defined in this way is called the arc price elasticity of demand. If we now assume Δp to be infinitesimal we get the concept of point price elasticity of demand (ε_d).

$$\varepsilon_d = \lim_{\Delta p \to 0} \frac{\Delta q}{\Delta p}.\frac{p}{q} = \frac{p}{q}.\lim_{\Delta p \to 0} \frac{\Delta q}{\Delta p} = \frac{p}{q}.\frac{dq}{dp} \qquad \ldots (10.3)$$

p and q are restricted to non-negative values only. Since the demand curve is generally downward sloping, i.e., $\frac{dq}{dp} < 0$, the price elasticity of demand is generally negative. The negative value of the right hand expression in (1) gives us absolute elasticity. Thus,

$$|\varepsilon_d| = -\frac{p}{q}.\frac{dq}{dp}$$

This can take any value between zero and infinite. The demand is said to be

i) perfectly inelastic $|\varepsilon_d| = 0$

ii) inelastic if $0 < |\varepsilon_d| < 1$

iii) unity elastic if $|\varepsilon_d| = 1$

iv) elastic if $1 < |\varepsilon_d| < \infty$

v) perfectly elastic if $|\varepsilon_d| \to \infty$

Note: Since the value of ε_d depends on initial price-quantity combination, there is no necessary connection between the slope and price elasticity of demand curve. Curve with different slopes may have the same elasticity and curves with different elasticities may have the same slope.

Now, suppose that price remains the same and income changes. The demand function in this case can be written as $q = f(y)$. The income elasticity of demand (η_d), measures the percentage change in quantity demanded due to one percent change in income (y).

Thus, $\eta_d = \dfrac{dq/q \times 100}{dy/y \times 100} = \dfrac{dq}{dy}.\dfrac{y}{q}$

In measuring the income elasticity of demand we assume price to be fixed at a certain level. The demand curve in this case can be written in the following alternative form $q = q(y)$. In this form, the demand curve is called the Engel curve. The income elasticity of demand is thus, the elasticity of the Engel curve.

The shape of the Engel curve depends on the nature of the commodity in question. If the good is a luxurious one its demand will increase with increasing income and the Engel curve slopes upward. Thus, for luxurious goods η_d is positive. Again for luxurious goods demand not only increases with increase in income, but it increases more than proportionately than increase in income. Thus,

$$\frac{dq}{q} > \frac{dy}{y} \rightarrow \frac{dq}{dy}\Big/\frac{q}{y} > 1 \rightarrow \eta_d > 1$$

For necessities $0 < \eta_d < 1$ and for inferior goods $\eta_d < 0$.

10.6.4 Constant Elasticity Demand Curve

The price elasticity of demand differs at different points on a demand curve. But we may have a special kind of demand curve every point of which shows same constant elasticity. The exponential demand functions are of this nature.

The general form of the constant elasticity demand curve is given by $q = Ap^{\alpha}$ where A, α are constants.

Let us derive the elasticity expression for this function at any point on this curve.

$$\varepsilon_d = \frac{p}{q}.\frac{dq}{dp} = \frac{p}{q}.\frac{d}{dp}(Ap^{\alpha}) = \frac{p}{q}.A\alpha.p^{\alpha-1} = \frac{p}{q}.\frac{q}{p}.Ap^{\alpha} = \alpha$$

Thus, ε_d is a pure number and constant throughout the whole curve. Consider a particular case when $\alpha = -1$. Then, $q = Ap^{-1} or,\ pq = A$.

In this case, $\frac{p}{q}.\frac{dq}{dp} = -1$. Such a term and curve is represented by rectangular hyperbola.

10.7 SLOPE AND CURVATURE

The first and second-order derivatives are of much use in the graphical analysis of economic functions. We know the first derivative of a function at any point measures the slope of the function at that particular point. Consider a function $y = f(x)$. The value of $f'(x)$ at $x = a$ is $f'(a)$. $f'(a) > 0$ implies that at $x = a$ the tangent gradient of the curve is positive, i.e., tangent to the curve at the curve $x = a$ slopes upwards. This implies that $f(x)$ increases through $x = a$

as x increases. Thus, the sign of the first order derivative informs us whether the function increases upwards or decreases downwards at the point in question. If $f'(x) > 0$ for all values of x in the domain of the function, then we can say $f(x)$ rise continuously from left to right.

The function which rises continuously is called monotonically increasing function. Thus, the function to be monotonically increasing implies the first derivative must be positive. Similarly, a function $y = f(x)$ is said to be monotonically decreasing if $f'(x) < 0$ for all x.

Example 10.7:

i) Consider the demand function $q = 3 - 5p$. Here $\frac{dq}{dp} = -5 < 0$. Thus, the function is a monotonically decreasing function.

ii) $c = aq^2 + bq + c$, $(a, b, c > 0)$. This cost function is monotonic because $\frac{dq}{dp} = 2aq + b > 0$ given $q \geq 0$. Actually, this function is monotonically increasing.

From the first derivative we learn about the direction of change of $f(x)$ with a change in x. The function $f(x)$ changes in the same direction at which x changes if $f'(x) > 0$ and in the opposite direction if $f'(x) < 0$.

One question immediately follows. What will be the pattern of change of $f(x)$ with a change in x? How can we know whether a function, which is known to be increasing, increases at a constant rate, or at an increasing rate or at a decreasing rate? The second derivative plays its role here. The bending of a curve or

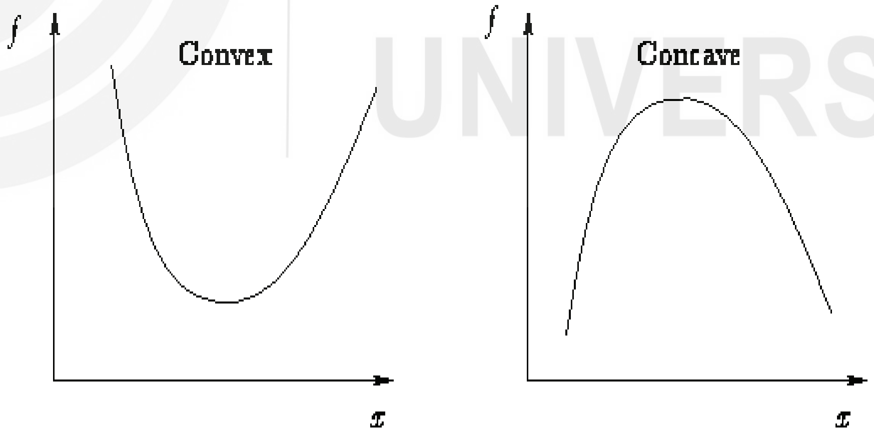

Fig. 10.2: Convex function and Concave function

curvature is measured by the rate of change of $f'(x)$ or the slope of the curve (Refer Fig.10.2). If the rate of change of the slope is positive, i.e.,$f''(x) > 0$, the curve is said to be convex. For a convex curve$f(x)$ increases or decreases at an increasing rate. If $f''(x) < 0$, the curve increases or decreases at a decreasing rate. Such curves are known as concave curves. Lastly, the curves which increase or decrease at a constant rate have $f''(x) = 0$, e.g., linear curves.

Below we summarise the nature of the function $y = f(x)$ depending on the values of $f(x)$ and $f''(x)$.

If, as x increases through a, then			
The curve at $x = a$ is			**The tangent to the curve at $x = a$**
$f'(a) > 0$ $f''(a) > 0$	upward rising convex	}	turns anti-clockwise
$f'(a) > 0$ $f''(a) = 0$	upward rising linear	}	does not turn
$f'(a) > 0$ $f''(a) < 0$	upward rising concave	}	turns clockwise
$f'(a) < 0$ $f''(a) > 0$	downward sloping convex	}	turns anti-clockwise
$f'(a) < 0$ $f''(a) = 0$	downward sloping linear	}	does not turn
$f'(a) < 0$ $f''(a) < 0$	downward sloping concave	}	turns clockwise

Example 10.8:

i) The function $y = 5x^2 + 6$ is convex because $f''(x) = 5 > 0$.

ii) Consider the demand function $p = aq^2 + bq + c$. Obtain the expression for its price elasticity of demand and tell what restriction should be imposed upon the value of q to make the demand function unitary elastic.

$$\varepsilon_d = \frac{p}{q}.\frac{dq}{dp} = \frac{p}{q}.\frac{1}{2aq+b} = \frac{aq^2+bq+c}{2aq^2+bq}$$

$$\text{If } \varepsilon_d = 1, \frac{aq^2+bq+c}{2aq^2+bq} = 1 \qquad \text{i.e., } aq^2 = c, \text{ i.e., } q = \sqrt{c/a}$$

iii) Demand function is $= aq^b, (a > 0, b > 0)$. Obtain the marginal revenue function and comment on its shape.

Since $R = p.q = aq^{b+1}$, $MR = \frac{dR}{dq} = a(b+1)q^b > 0$; for $q > 0$. Thus, MR is for all $q > 0$. The MR curve in this case is upward rising as $\frac{d^2R}{dq^2} = ab(b+1)q^{b-1} > 0$.

10.8 TAYLOR SERIES

Suppose, we wish to approximate a function $f(x)$ at some arbitrary point $x = a$ by a polynomial of the form -

$f_n(x) = a_o + a_1(x-a) + a_2(x-a)^2 + ... + a_n(x-a)^n$, where, $f_n(x)$ is the approximate value of $f(x)$.

This exercise will be particularly helpful in finding the conditions for extrema of functions. The functions expressed in the above form are called *analytic functions*.

Our next task is to find appropriate values of the coefficients of the above polynomial. To do this, we proceed as follows:

First, we note that $f(x)$ should equal $f_n(x)$ when $x = a$. Thus, $a_o = f(a)$.

To approximate $f(x)$ even better, let us make the derivatives of $f(x)$ and $f_n(x)$ equal at $x = a$. We know,

$$f_n'(x) = a_1 + 2a_2(x-a) + ... + na_n(x-a)^{n-1}$$

$$f_n''(x) = 2a_2 + 3.2.a_3(x-a) ..... + n(n-1)a_n(x-a)^{n-1}$$

....

$$f_n^n(x) = n!a_n$$

Clearly, when $x = a$, we get

$$a_1 = f'(a)$$

$$a_2 = \frac{f''(a)}{2!}$$

…..

Thus, we can write,

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + ...$$

This is known as Taylor series.

Note that we have not included the last term (i.e., the $(n+1)^{th}$ term) $\frac{f^n(a)}{n!}(x-a)^n$ in the above expression. This is because $f(x)$ may be *any* type of function. If we take the last term in the above form, then the approximation will not be exact if $f(x)$ itself is not a polynomial.

The power series can be made exact (when $f^{(n-1)}$ is continuous) if the last term is evaluated not at $x = a$, but at some point x^*, such that $x < x^* < a$ (why this is so is beyond the scope of the treatment).